

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: SUB-STRAND 2.4 AREA OF POLYGONS – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How can we calculate the exact area of any polygon-shaped surface, even when we cannot measure its height, so that builders, farmers and designers in our community never order too much or too little material? Use what you learned in the Kericho community hall courtyard unit to answer each prompt. Write in full sentences, show every step of your working, include units, and give reasons for your choices. Round areas to 1 decimal place unless told otherwise. You may use a calculator and your Courtyard Model.

Part 1: The Courtyard Puzzle and the Sine-Area Formula

A contractor in Kericho County must order tiles for a community hall courtyard. Some triangular sections have two known side lengths and the angle between them, but no height can be measured and there is no right angle. (a) Explain why $A = \frac{1}{2} \times \text{base} \times \text{height}$ cannot be used directly on these triangles. (b) Using a right-angled sub-triangle, show how the height can be written using side b and the included angle C. (c) Substitute this into $A = \frac{1}{2} \times \text{base} \times \text{height}$ to get a formula for the area. (d) Explain how this formula helps the contractor with Triangle A and Triangle B of the courtyard.

(a) The formula needs a perpendicular height, but the courtyard sections have no right angle and no height can be physically measured, so we cannot get a number to put in for height. (b) Drop a perpendicular from vertex A to side BC. This makes a right-angled triangle whose hypotenuse is side b and whose angle at C is the included angle. The height h is the side opposite angle C, so $\sin C = h/b$, which gives $h = b \sin C$. (c) Take the base as a, so $A = \frac{1}{2} \times a \times h = \frac{1}{2} \times a \times (b \sin C)$, which gives $A = \frac{1}{2}ab \sin C$. (d) The contractor measures two sides and the included angle with a tape and protractor and calculates the area without measuring any height. Using this formula, Triangle A is about 39.9 m² and Triangle B is about 21.8 m². The sine of the included angle carries the height information, so the missing height is no longer a problem.}

Part 2: Triangles with Three Sides Only (Heron's Formula)

Mwangi, a farmer near Eldoret in the Rift Valley, has a

(a) $\frac{1}{2}ab \sin C$ needs an included angle, and Mwangi has only the three side lengths and no angle, so another formula that uses sides alone is needed. (b) Heron's Formula is $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$, where $s = (a+b+c)/2$ is the semi-perimeter, half the

triangular maize plot with sides 40 m, 55 m and 63 m. He cannot measure any angle or any height. (a) Explain why $\frac{1}{2}ab \sin C$ is not the best choice for his plot. (b) Write down Heron's Formula and say what s stands for. (c) Calculate the area of the plot, showing each step. (d) A classmate says the answer will change if you call the 63 m side a instead of c. Do you agree? Explain using what you found in class.	perimeter. (c) $s = (40+55+63)/2 = 158/2 = 79$ m. Then $s-a = 39$, $s-b = 24$, $s-c = 16$. Area = $\sqrt{(79 \times 39 \times 24 \times 16)} = \sqrt{1,183,104} \approx 1,087.7$ m². (d) I disagree. Heron's Formula is symmetric: the terms $s-a$, $s-b$ and $s-c$ are multiplied together, so swapping which side is called a, b or c only changes the order of the multiplication and not the answer. In our group measurements, swapping labels gave the same area. So the farmer, or the contractor, can measure the sides in any order.}
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Part 3: The Parallelogram Stage – Choosing the Right Method

The stage in the Kericho courtyard is a parallelogram with adjacent sides 6 m and 4 m and an included angle of 65°. (a) Write the formula you would use and explain why it is the practical choice here. (b) Calculate the area of the stage. (c) Explain how this formula is connected to the triangle formula $\frac{1}{2}ab \sin C$. (d) A kanga patch is a rhombus whose two diagonals can both be measured. Which formula would you choose for it, and why? (e) The contractor needs whole tiles. Should the number of tiles be rounded up or down? Why?	(a) Area = $ab \sin \theta$. It is practical because the courtyard stage has no accessible perpendicular height, but two side lengths and the included angle can be measured with a tape and protractor. (b) Area = $6 \times 4 \times \sin 65^\circ = 24 \times 0.9063 \approx 21.75$ m², about 21.8 m². (c) A diagonal splits a parallelogram into two congruent triangles. Each has area $\frac{1}{2}ab \sin \theta$, so the parallelogram has twice that: $2 \times \frac{1}{2}ab \sin \theta = ab \sin \theta$. (d) I would use $A = \frac{1}{2}d_1d_2$. When both diagonals are measurable, it is quick, because the diagonals of a rhombus cross at right angles and split it into four right-angled triangles. Rule: use the diagonal method when diagonals can be measured, and the sine method when sides and an included angle are known. (e) Round up. If the number is rounded down, there will not be enough tiles to cover the whole surface, so the contractor would be under-ordering material.}
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Part 4: Regular Pentagons and Hexagons from Side Length Alone

(a) A njugu (groundnut) vendor at Kericho market uses a regular pentagonal tray with side length s. Explain, step by step, how dividing the pentagon	(a) Joining the centre to the five vertices splits the pentagon into 5 congruent isosceles triangles. Each has an apex angle of $360^\circ \div 5 = 72^\circ$ and base angles of 54°. The perpendicular from the centre to the midpoint of a side (the apothem) bisects the apex angle, so a right-angled triangle forms with a 36° angle at the centre and a base of $s/2$. Then $\tan 36^\circ = (s/2) \div \text{apothem}$, so apothem = $(s/2)/\tan 36^\circ$. Area of one triangle = $\frac{1}{2} \times s \times (s/2)/\tan 36^\circ = s^2/(4 \tan 36^\circ)$. Multiplying by 5 gives $A = 5s^2/(4 \tan 36^\circ)$. $\tan 36^\circ$ appears because the right-angled half-triangle has an angle of 36°, which is half of 72°. (b) $A = 5 \times 9^2 \div (4 \times \tan 36^\circ) = 405 \div 2.906$
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into congruent triangles leads to $A = 5s^2/(4 \tan 36^\circ)$. Include the angles in one triangle and say why $\tan 36^\circ$ appears. (b) Use the formula to find the area of a tray with $s = 9$ cm. (c) The hexagonal border tiles of the courtyard are regular hexagons with known side length s. Explain how to derive their area formula from the area of an equilateral triangle and say why no height measurement is needed.	$\approx 139.4 \text{ cm}^2$. (c) A regular hexagon is made of 6 identical equilateral triangles. An equilateral triangle of side s has area $\frac{1}{2} \times s \times s \times \sin 60^\circ = (\sqrt{3}/4)s^2$, which is the sine-area formula specialised to an equilateral triangle. Multiplying by 6 gives $A = 6 \times (\sqrt{3}/4)s^2 = (3\sqrt{3}/2)s^2$. It uses only the side length s, so the contractor does not need to measure a height, and the hexagon formula comes from the same family as the triangle formula.}
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Part 5: Putting It All Together for the Kericho Courtyard

Your final task is to explain the answer to the driving question. (a) For each courtyard section, say which formula you would use and what measurements the contractor needs: (i) a triangle with two sides and the included angle; (ii) a triangle with three sides; (iii) the parallelogram stage; (iv) a hexagonal border tile. (b) Explain why the Lesson 1 approach of base $\times$ height only could not give the total area. (c) Explain how knowing every section's area helps the contractor avoid ordering too much or too little material.	(a) (i) Area = $\frac{1}{2}ab \sin C$, with two side lengths and the included angle. (ii) Heron's Formula, Area = $\sqrt{s(s-a)(s-b)(s-c)}$ with $s = (a+b+c)/2$, with all three side lengths. (iii) Area = $ab \sin \theta$, with two adjacent sides and the included angle. (iv) Area = $(3\sqrt{3}/2)s^2$, with the tile side length s only. (b) Base $\times$ height needs a perpendicular height. In the courtyard no heights can be measured and there are no right angles, so it fails for the triangles, the stage and the hexagonal tiles. (c) Each formula uses only measurements that can be taken on site: side lengths, an included angle, or tile side. So the contractor can find the area of every section, add them to get the total area, and divide by the area of one tile. He rounds tile counts up so he does not under-order, and because the areas are accurate he does not order a large surplus. In this way the formulas let builders, farmers and designers in our community buy the right amount of material without ever dropping a perpendicular.}
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Choosing and justifying the correct formula	Selects the correct formula for every shape and explains clearly, using the available measurements, why it is the practical choice when no height can be	Selects the correct formula for most shapes and gives a reasonable but incomplete reason.	Chooses formulas incorrectly or without giving a reason.

	measured.		
Derivations and links between formulas	Shows correctly how $h = b \sin C$ leads to $\frac{1}{2}ab \sin C$, and links the parallelogram, equilateral triangle, hexagon and pentagon formulas to the triangle formulas with clear steps and angles.	Gives most steps of the derivations, with a small gap in the angles or in the link between formulas.	States formulas without showing how they are derived or connected.
Accurate calculation	Calculations are correct and fully shown (Heron's $\approx 1,087.7 \text{ m}^2$, stage $\approx 21.75 \text{ m}^2$, pentagon $\approx 139.4 \text{ cm}^2$), with units and sensible rounding.	Method is correct but with a minor arithmetic, rounding or unit error.	Method is incorrect or the working is not shown.
Connecting to the Kericho courtyard and driving question	Clearly explains how the formulas let the contractor find every area from measurable data and avoid over- or under-ordering, including rounding up tile counts.	Links the formulas to the courtyard but gives limited detail on ordering material.	Gives little or no link to the courtyard or the driving question.

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